

Knowing When and Why to Refuse: Defect-Aware Explanation and Recovery for Assistive Visual Question Answering

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ABSTRACT Visual question answering (VQA) systems for blind and low-vision users must do more than return an answer: they must know when an answer is unreliable, explain why the interaction failed, and help the user recover. This article studies a lightweight reliability layer for assistive VQA that combines answerability triage, human-labeled image-quality defect diagnosis, selective-prediction diagnostics, and actionable retake guidance. Using VizWiz-VQA and VizWiz-QualityIssues, real answerability labels are joined with seven human-annotated quality defects: blur, bright, dark, obstruction, framing, rotation, and unrecognizable content. Small prediction heads trained on frozen CLIP, DINOv2, and MobileNetV3 embeddings estimate answerability and defects, while frozen ViLT and BLIP VQA models provide answers and confidence scores. The strongest triage head reaches AUROC 0.7926 with CLIP; the strongest defect head reaches mAP 0.6309 with DINOv2; and CLIP obtains ungated actionable recovery 0.7196. When retake guidance is gated by refusal at a 90% target coverage, CLIP reaches gated actionable recovery 0.875 with gated false refilm rate 0.050 under the ViLT gate. Selective prediction is model-dependent: auxiliary reliability signals do not significantly improve ViLT confidence after false-discovery-rate correction, but they do improve the weaker BLIP confidence signal, with full-stack learned risk models improving BLIP AURC by up to 0.0179. Defect diagnosis is therefore best viewed not as a universal replacement for confidence-based risk ranking, but as an interpretable refusal-and-recovery mechanism whose ranking value depends on the underlying VQA model.

INDEX TERMS Accessibility, assistive technology, calibration, image quality assessment, selective prediction, visual question answering.

I. INTRODUCTION

VISUAL question answering (VQA) is increasingly framed as an assistive interface: a blind or low-vision user captures an image, asks a spoken question, and expects a task-relevant answer. VizWiz made this promise concrete by collecting images and questions from blind users in everyday settings, and in doing so exposed a central obstacle for deployment: the visual evidence is often incomplete, poorly framed, blurred, dark, bright, rotated, or simply insufficient to answer the question at hand [1]. Subsequent work around the VizWiz ecosystem has reinforced that this difficulty is structural rather than incidental. Answer grounding remains hard in images taken by people with visual impairments, salient objects in those images differ systematically from those in

conventional benchmarks, assistive captioning systems still struggle to transfer from curated data, and even newer ego-centric video-based benchmarks remain challenging despite richer visual context [2]–[5]. In other words, the practical problem in assistive VQA is not only to answer correctly when the image is good enough, but also to recognize when the image–question pair is unsafe to trust, which is precisely the deployment setting our reliability layer is designed for.

A substantial literature now studies this reliability problem through selective prediction and abstention. Reliable VQA work showed that a system should sometimes prefer saying “I do not know” to returning a confident error, and that learned selection functions can improve the risk–coverage trade-off over naive confidence heuristics [6], [7]. More recent work

extends this agenda to broader vision-language reasoning, including selectively answering visual questions, identifying unreliable black-box vision-language outputs through consistency and uncertainty cues, and reducing unnecessary abstention once a model has become too conservative [8]–[10]. That literature is essential because it clarifies how to rank answer risk, but it usually stops at the decision to answer or abstain. For assistive use, however, refusal alone is not enough: a user also needs a reason for the failure and guidance on how to recover, which is the part of the problem our paper brings into the foreground.

A parallel thread of work studies image quality in assistive imagery more directly. VizWiz-QualityIssues showed that image-quality assessment for blind-photographer images must move beyond abstract perceptual scores and instead identify concrete downstream-relevant defects, such as blur, poor lighting, obstruction, framing problems, and unrecognizable content [11]. User-centered studies tell a similar story from the interaction side: blind users actively try to diagnose recognition failures by changing viewpoint, distance, background, and framing, yet error recovery remains effortful and unreliable without explicit support from the system itself [12]. At the same time, prior assistive-vision work has typically isolated one piece of the loop—answerability, quality prediction, grounding, captioning, or user behavior—rather than integrating when to refuse, why to refuse, and how to guide a re-shot inside one reliability pipeline. Our contribution is to connect these pieces by treating defect diagnosis not as an auxiliary benchmark alone, but as an explanatory and recovery-oriented layer for assistive VQA.

A further challenge is that not all confidence signals mean the same thing. In discriminative models, maximum-softmax confidence is easy to extract and, while often miscalibrated, has a well-established post-hoc calibration literature [13]. In autoregressive generation, by contrast, raw sequence probability is a less stable proxy for correctness: recent work shows that sequence likelihood can mis-rank candidate quality and often benefits from explicit recalibration or context-sensitive reweighting before it can serve as a dependable confidence measure [14], [15]. This distinction matters directly for assistive VQA because modern systems increasingly mix discriminative and generative answerers whose confidence semantics are not interchangeable. Our paper therefore asks a narrower and more deployment-relevant question than whether auxiliary signals “help” in the abstract: when does answerability and defect information actually improve selective prediction beyond the base VQA confidence, and when is its main value instead to explain refusals and guide recovery?

We answer this question by building a lightweight reliability layer around frozen VQA models rather than proposing a new answerer. The layer predicts answerability and image-quality defects from frozen visual embeddings, harvests VQA confidence from a frozen answerer, evaluates selective-prediction behavior, and maps predicted defects to retake guidance. This design lets us separate risk ranking from user-facing diagnosis, and it yields four contributions:

- We join VizWiz-VQA answerability labels with VizWiz-QualityIssues defect labels to study answerability, defect diagnosis, refusal, and recovery inside one assistive VQA pipeline.
- We evaluate lightweight answerability and multi-label defect heads on frozen CLIP, DINOv2, and MobileNetV3 embeddings, keeping the reliability layer modular and computationally lightweight.
- We introduce the Actionable Recovery Rate (ARR) and False Refilm Rate (FRR) to evaluate whether predicted defects translate into useful retake guidance, and we report both ungated diagnostics and refusal-gated deployment-aligned rates.
- We show that selective-prediction gains from auxiliary signals are model-dependent: they do not significantly improve the stronger ViLT confidence ranking after correction, but they do improve the weaker BLIP confidence signal, while defect diagnosis remains valuable in both settings as an interpretable refusal-and-recovery mechanism.

II. RELATED WORK

A. ASSISTIVE VISUAL QUESTION ANSWERING AND THE VIZWIZ LINEAGE

Assistive visual question answering is motivated by a real accessibility setting rather than a closed benchmark game. VizWiz established that distinction by collecting images and spoken questions from blind users, thereby surfacing three properties that are far less prominent in curated VQA benchmarks: conversational questions, poor image capture conditions, and a substantial prevalence of unanswerable examples [1]. More recent accessibility-oriented work has extended this line of inquiry in two complementary directions. Context-VQA argues that users who cannot see an image often need answers that are conditioned not only on image content but also on the surrounding use context, such as whether an image appears in shopping, travel, or news settings [16]. VizWiz-LF then shows that blind and low-vision users often benefit from richer answers containing explanation and suggestion, while also documenting that generated long-form answers are especially vulnerable to hallucinating details when the image is blurry, irrelevant, or otherwise unanswerable [17]. Taken together, these studies show that assistive VQA requires systems that can recognize failure modes in realistic user images rather than merely maximize answer accuracy, which is precisely the setting in which our work introduces a separate reliability layer around a frozen VQA answerer.

B. IMAGE QUALITY ASSESSMENT FOR REAL-WORLD ASSISTIVE IMAGES

A second line of work studies why images captured in the wild fail downstream vision tasks. VizWiz-QualityIssues is central here because it links images taken by blind users to human judgments about recognizability and concrete defect categories such as blur, framing, brightness, darkness,

obstruction, and rotation [11]. In the broader no-reference image quality assessment literature, recent methods increasingly emphasize authentic, in-the-wild degradations rather than synthetic distortions, reflecting the fact that real deployment images contain mixed technical and semantic failures that are harder to score with a single scalar label [18]. Work on visually impaired user-generated content pushes this further: Mandal et al. build a large perceptual dataset of low-quality images captured by visually impaired users and propose automated feedback aimed at helping users mitigate common technical distortions when retaking photographs [19]. Our paper builds on this trajectory but shifts the unit of analysis from generic visual quality to task-facing defect diagnosis for assistive VQA, using human-labeled defect signals not only to predict quality problems but also to explain refusals and drive recovery guidance.

C. SELECTIVE PREDICTION AND ABSTENTION IN VQA

Selective prediction provides the formal language for saying “answer when reliable, abstain when risky” and is therefore the natural starting point for reliable assistive VQA [20]. In VQA specifically, Whitehead et al. showed that naive confidence thresholding can be too brittle and that learned selection mechanisms can improve the coverage–risk trade-off, while Dancette et al. demonstrated that stronger selectors can be learned by borrowing information from peer models [6], [7]. More recent work extends selective answering to contemporary vision-language models: Srinivasan et al. study how to reduce needless abstention in vision-language reasoning, and Khan and Fu analyze black-box uncertainty and consistency signals for detecting unreliable vision-language model (VLM) responses [9], [10]. This literature mainly treats abstention as a ranking problem over answer risk; our work is complementary in asking when auxiliary answerability and defect signals improve that ranking, and in separating the ranking function from the user-facing need to explain a refusal and recommend a corrective retake action.

D. CONFIDENCE CALIBRATION AND UNCERTAINTY IN VISION-LANGUAGE MODELS

Confidence quality is a crucial variable in any selective-prediction pipeline, because auxiliary signals should only help when the base confidence signal leaves meaningful headroom. For discriminative classifiers, modern calibration work has made temperature scaling the standard post-hoc reference point for repairing overconfident softmax outputs without changing class order [13]. Autoregressive generation is more complicated. In neural machine translation, Wang et al. show that severe miscalibration can persist at inference time even when training-time behavior appears better behaved [21]. In natural-language generation more broadly, Lin et al. argue that raw generated-sequence likelihood is a noisy confidence surrogate because it conflates semantic correctness with syntactic and phrasing effects, and they show that reweighting sequence likelihood can substantially improve reliability as a correctness signal [15]. Recent multi-

modal studies report a similar pattern at the vision-language level, finding persistent miscalibration and a tendency for multimodal large language models to answer rather than admit uncertainty [22]. We therefore position our ViLT–BLIP comparison not as a universal claim that generative confidence is always inferior to discriminative softmax, but as a targeted empirical test of whether auxiliary defect and triage signals become more valuable when the underlying confidence channel is weaker or less well aligned with correctness.

E. ACTIONABLE FEEDBACK AND CAMERA GUIDANCE FOR BLIND USERS

The final gap concerns actionability. Many assistive systems can produce an answer, a caption, or a confidence score, yet far fewer tell the user how to recover from a failed capture. Mandal et al. come closest by proposing automated feedback to help visually impaired users take better photographs, but their emphasis is primarily on mitigating technical distortion rather than on tying those signals to question answerability and refusal decisions [19]. Conversely, long-form assistive VQA work shows that explanations and suggestions are desirable in principle, while also demonstrating how easily generative systems can hallucinate when the underlying image is not dependable [17]. What remains largely missing is a deployment-oriented evaluation of whether a refusal leads to a useful corrective action or merely increases user burden. Our contribution targets that gap directly by turning predicted defect categories into deterministic retake guidance and by evaluating the result with actionable recovery metrics, ARR and FRR, including refusal-gated variants that better match real assistive deployment.

III. DATA CONSTRUCTION AND SPLIT PROTOCOL

A. SOURCE DATASETS

We build the study around two complementary VizWiz resources. VizWiz-VQA provides real images captured by blind users together with spoken questions, ten human answers, and an answerability flag indicating whether the question can be answered from the image [1]. VizWiz-QualityIssues provides human annotations for image-quality defects in the same assistive-photography setting, including blur, bright, dark, obstruction, framing, rotation, and unrecognizable content [11]. Using these two resources together lets us study not only whether a system should answer, but also whether it can explain the likely visual failure mode when it should not.

B. JOINING LABELS AND ENFORCING THE FROZEN-KNOB PROTOCOL

The two datasets are joined by image filename and split, and we retain only examples that have both VQA answerability labels and quality-issue annotations. This produces 24,319 rows in total, comprising 20,000 training rows and 4,319 validation rows. We further divide the validation portion into a calibration subset of 1,295 rows and a report subset of 3,024

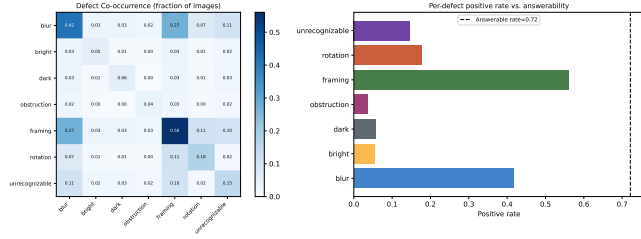


FIGURE 1. Quality-label structure in the joined data. The left panel shows pairwise defect co-occurrence, while the right panel shows per-defect prevalence relative to the answerable rate. The skewed distribution motivates per-defect metrics and recovery analysis rather than aggregate accuracy alone.

rows. This split is central to the paper’s evaluation discipline: all operating thresholds, temperature-scaling parameters, and learned selective-prediction risk models are fit only on the calibration subset, and all numbers reported in the paper are computed on the held-out report subset. This frozen-knob rule prevents threshold tuning on the same examples used for final reporting and keeps the reliability analysis deployment-aligned rather than retrospectively optimized.

C. LABEL DEFINITION AND CLASS IMBALANCE

For quality defects, we follow the VizWiz-QualityIssues annotation protocol: each image is labeled by five crowdworkers, and a defect is treated as present when at least two annotators select it [11]. After joining, the answerable rate is 0.7214. The defect labels are substantially imbalanced. Framing and blur are common, with positive rates 0.5606 and 0.4161, respectively, while bright, dark, and obstruction are much rarer, with positive rates 0.0539, 0.0563, and 0.0352. These skews matter for evaluation: they make aggregate accuracy misleading, motivate per-defect AUROC and AUPRC reporting, and make recovery analysis more informative than a single pooled score.

Figure 1 illustrates the structure of the joined labels. The co-occurrence matrix shows that defects are not independent: framing often appears with blur and unrecognizable content, for example, which means that a single failed assistive-VQA interaction may admit several plausible visual explanations. The prevalence plot also makes clear why answerability and quality diagnosis should be separated conceptually. An image may still be answerable despite containing a real defect, and conversely an unanswerable image may fail for reasons that are only partially captured by any one defect label. This joined dataset therefore supports the paper’s central decomposition into answerability triage, defect diagnosis, refusal, and recovery rather than collapsing all failures into one binary notion of correctness.

IV. EVALUATION METRICS

A. ANSWERABILITY AND DEFECT PREDICTION

Because the joined VizWiz labels are substantially imbalanced, we separate threshold-free ranking metrics from thresholded operating-point metrics. For answerability triage,

we report AUROC and AUPRC to measure how well the score orders answerable and unanswerable examples across all thresholds. We then report F1 and balanced accuracy at the calibration-selected operating point. This combination is deliberate: AUPRC is especially informative when positives and negatives are skewed, whereas balanced accuracy prevents the majority answerability class from dominating the summary [23]. We also keep a majority-class baseline to show whether a learned triage head improves on a trivial deployment rule. This metric bundle directly supports our contribution by evaluating, separately and honestly, whether the reliability layer can screen answerability under the class imbalance that characterizes assistive imagery.

For defect diagnosis, we treat the task as seven-label multi-label prediction and report per-defect AUROC and AUPRC, together with macro-F1, micro-F1, and mean average precision (mAP). Let $L = 7$ denote the number of defect labels and AP_ℓ the average precision for label ℓ ; then

$$\text{mAP} = \frac{1}{L} \sum_{\ell=1}^L AP_\ell. \quad (1)$$

Per-defect reporting matters because blur, framing, and unrecognizable content behave very differently from rare defects such as bright, dark, and obstruction, and aggregate scores can otherwise make a detector look more uniform than it really is. Reporting both macro- and micro-averaged summaries therefore lets us separate overall usefulness from minority-label behavior, which is exactly the distinction needed when defect labels are used not only for prediction but also for explanation and recovery.

B. SELECTIVE PREDICTION

We evaluate answer-versus-refuse behavior through the selective-prediction lens of risk versus coverage [20], [24], [25]. After sorting report-set examples by decreasing trust score $s_{(1)} \geq s_{(2)} \geq \dots \geq s_{(n)}$, where higher values mean “more willing to answer,” we define the empirical selective risk at retained prefix length k as

$$R_k = \frac{1}{k} \sum_{j=1}^k (1 - a_{(j)}), \quad (2)$$

where $a_{(j)} \in [0, 1]$ is the VizWiz VQA accuracy of the retained example. Coverage is k/n , and the area under the empirical risk-coverage curve is

$$\text{AURC} = \frac{1}{n} \sum_{k=1}^n R_k. \quad (3)$$

Lower AURC is better because it means the scoring rule pushes errors later in the answered queue and therefore leaves lower risk on higher-coverage retained sets [6]–[10]. This is the right headline metric for our paper because the deployment decision we study is answer versus refuse, not calibration in isolation.

Calibration is still relevant, but we use it as an auxiliary diagnostic rather than a substitute for risk ranking. We compare raw global VQA confidence with temperature-scaled confidence [13], and when calibration behavior is discussed we report expected calibration error (ECE),

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad (4)$$

where B_b is confidence bin b . We interpret ECE cautiously because its value depends on binning and related design choices, a point emphasized in later calibration studies [26], [27]. Against this background, we compare the following score families for each frozen VQA model: global confidence, temperature-scaled confidence, defect-conditioned calibration, learned risk scores combining confidence with predicted defects, triage score alone, learned confidence-plus-triage scores, learned full-stack scores using confidence, triage, and predicted defects, and a confidence-plus-ground-truth-defects model used only as an oracle diagnostic of defect information. This comparison grid is what allows us to make the paper's central model-dependent claim without assuming in advance that defect signals must beat confidence.

C. ACTIONABLE RECOVERY

Standard abstention metrics stop at the decision to answer or refuse, but assistive use requires a user-facing next action. We therefore evaluate retake guidance with two complementary measures. Let U be the set of unanswerable examples and A the set of answerable examples. For image i , let $\hat{d}_i$ be the top-scoring predicted defect if its probability exceeds 0.5, and let D_i be the set of ground-truth defects. We define the Actionable Recovery Rate as

$$\text{ARR} = \frac{1}{|U|} \sum_{i \in U} \mathbb{1}[\hat{d}_i \in D_i], \quad (5)$$

so ARR measures whether the chosen retake recommendation corresponds to at least one true failure mode on unanswerable images. We define the ungated False Refilm Rate as

$$\text{FRR} = \frac{1}{|A|} \sum_{i \in A} \mathbb{1}[\hat{d}_i \neq \emptyset], \quad (6)$$

which counts how often the system would ask for a retake on images that are, in principle, answerable. This construction is intentionally conservative: an answerable image may still contain a genuine defect, so FRR is a burden proxy rather than a pure error rate.

We also report refusal-gated variants of ARR and FRR at target coverages of 90%, 80%, and 70%, where retake guidance is issued only when the VQA confidence falls below a calibration-selected refusal threshold. This distinction is important in practice because ungated defect prediction answers the diagnostic question “what is wrong with this image?”, whereas gated recovery answers the deployment question “when should the system actually ask the user to re-shoot?” The gated metrics therefore make our recovery

analysis deployment-aligned: they evaluate whether a refusal can lead to a useful next action rather than stopping at abstention.

D. JOINT VERSUS CASCADE

To quantify architectural effects, we compare a cascade design with separate answerability and defect heads against a joint multi-task head, and we summarize the difference as the change in answerability AUROC between the two designs. This comparison is intentionally narrow: the goal is not to claim a new multi-task learning result, but to determine whether head coupling materially changes triage quality once the visual backbone is frozen. That framing keeps the architectural comparison subordinate to the paper's main question, namely whether a lightweight reliability layer can provide trustworthy refusal, explanation, and recovery without retraining the VQA model.

Across selective-prediction comparisons, uncertainty is estimated with paired bootstrap confidence intervals on the held-out report subset, following the standard resampling logic that avoids strong parametric assumptions for complex metrics [28]. Because we evaluate families of closely related hypotheses on the same report split, we additionally apply Benjamini–Hochberg false-discovery-rate correction within each family rather than interpreting nominal p -values one by one [29]. This conservative testing protocol is essential to our contribution because the paper's main claim is deliberately conditional: it is just as important to show where auxiliary signals do not improve selective prediction as where they do.

V. METHODOLOGY

Figure 2 summarizes the proposed reliability layer. The central design choice is role separation. A frozen visual encoder transforms each image into a reusable embedding, and lightweight prediction heads operating on that embedding estimate two auxiliary properties: whether the image–question pair is answerable and which image-quality defects are likely to be present. In parallel, a frozen visual question answering (VQA) model produces an answer and an associated confidence score. These signals support three related but distinct operations: ranking answer risk, explaining why an interaction may have failed, and recommending how the user might recover.

The proposed system does not retrain the underlying VQA model or claim to improve its answer-generation capability. Instead, it places a lightweight reliability layer around a frozen answerer. This modular construction allows us to test confidence, answerability triage, predicted defects, and their learned combinations without changing the base VQA model. It also supports the paper's central distinction: VQA confidence may rank answer risk, whereas defect diagnosis can explain a refusal and guide a corrective retake.

A. FROZEN VISUAL BACKBONES

We compare three frozen image encoders: CLIP ViT-B/32 [30], DINOv2 ViT-S/14 [31], and MobileNetV3-Large [32].



FIGURE 2. Overview of the proposed reliability layer. Frozen visual features feed the answerability and defect-prediction heads, while the frozen VQA model supplies an answer and a confidence score. Confidence supports risk ranking and the answer/refuse decision; predicted defects explain refusals and generate retake guidance.

Each backbone produces a fixed visual representation for an input image. The resulting embeddings are cached once and reused by all downstream heads, avoiding repeated feature extraction and ensuring that the answerability and defect experiments operate on identical representations.

The selected backbones represent different visual-learning strategies. CLIP provides a language-supervised visual representation, DINOv2 provides a self-supervised transformer representation, and MobileNetV3 provides a comparatively lightweight convolutional representation. Because all encoders remain frozen, observed differences reflect the information available in their representations and the behavior of the lightweight heads rather than end-to-end backbone fine-tuning.

B. ANSWERABILITY TRIAGE

For each backbone, a lightweight multilayer perceptron predicts the VizWiz answerability label. The target represents whether the question can be answered from the available visual evidence; it does not directly represent whether the frozen VQA model will produce the correct answer. This distinction is necessary because an answerable example may still be answered incorrectly, while an image containing one or more quality defects may remain answerable.

The heads are trained on the training split using five random seeds. Decision thresholds are selected exclusively on the calibration subset, after which the selected thresholds are frozen. Final performance is measured on the held-out report subset and reported as the mean and standard deviation across seeds. The same triage scores are later evaluated both independently and as auxiliary inputs to learned selective-prediction models.

C. DEFECT DIAGNOSIS

The defect head is formulated as a multi-label classifier over seven image-quality conditions: blur, bright, dark, obstruction, framing, rotation, and unrecognizable content. A multi-label formulation is required because a single assistive photograph can contain several simultaneous defects. For example, an image can be both blurred and poorly framed, or both dark and partially obstructed. Each defect is therefore assigned an independent prediction probability.

The defect head has two primary user-facing roles. First, it provides a possible explanation when the system refuses to answer. Second, the predicted defect is mapped to a

corresponding retake instruction. The same continuous defect probabilities are also evaluated as possible auxiliary features for selective prediction. However, whether they improve risk ranking is treated as an empirical question rather than assumed by the architecture.

D. FROZEN VQA ANSWERERS AND CONFIDENCE HARVESTING

We use two frozen VQA answerers with different output and confidence structures. The first is ViLT [33], which produces a predicted answer through a discriminative answer classifier. Its confidence is the maximum probability in the model's softmax answer distribution. The second is BLIP-VQA-base [34], which generates an answer autoregressively. Its confidence is the length-normalized sequence probability, calculated as the geometric mean of the probabilities assigned to the generated tokens.

Using both answerers enables a controlled comparison between a discriminative maximum-softmax score and a generative sequence-probability score. Neither answerer is fine-tuned during the present study. Their predictions and confidence values are harvested as fixed outputs so that the analysis isolates the contribution of the surrounding reliability layer.

VizWiz answer accuracy is computed as

$$\text{Acc}(y, \mathcal{A}) = \min\left(\frac{m}{3}, 1\right), \quad (7)$$

where m is the number of exact matches between the model prediction y and the ten human reference answers $\mathcal{A}$. This consensus score supplies the correctness target used in the selective-prediction analysis.

E. SELECTIVE-PREDICTION FEATURE CONSTRUCTION

The VQA model's native confidence serves as the baseline risk-ranking signal. We then examine whether auxiliary reliability signals provide useful information beyond that confidence. The candidate auxiliary inputs are the answerability score produced by the triage head and the seven continuous probabilities produced by the defect head.

For each VQA answerer, the evaluated alternatives include native global confidence, temperature-scaled confidence, defect-conditioned calibration, confidence combined with predicted defects, triage score alone, confidence combined with triage, and a full-stack model combining confidence, triage, and predicted-defect probabilities. We additionally evaluate confidence combined with ground-truth defects as an oracle diagnostic. The oracle is not a deployable method; it tests whether perfect binary defect labels would provide an inherently better ordering of answer risk.

All temperatures, refusal thresholds, and learned risk models are fit exclusively on the calibration subset. They are then frozen and applied without modification to the report subset. This frozen-knob protocol prevents the final risk-coverage results from being influenced by post hoc tuning on the reported examples.

F. RECOVERY MAPPING AND GUIDANCE POLICY

Each predicted defect maps to a fixed user-facing retake action. For example, blur maps to holding the camera steady and refocusing, dark maps to adding light, and framing maps to stepping back so that the complete object is visible. The mapping is deterministic so that recovery success can be traced directly to the accuracy of the predicted defect rather than to the behavior of a separate language-generation model.

The highest-scoring defect is selected when its probability exceeds the prescribed threshold. On an unanswerable example, the guidance is considered actionable when the selected defect is present among the ground-truth defects. A retake suggestion issued for an answerable example contributes to false refile burden. These two outcomes are measured using ARR and FRR, respectively.

The deployment-aligned policy does not issue guidance whenever the defect head detects a quality issue. Instead, the confidence gate first decides whether the VQA answer should be accepted or refused. Defect-based guidance is presented only after refusal. This ordering is important because many answerable images still contain genuine quality defects; unconditional defect guidance would therefore create unnecessary retake requests even when the available visual evidence is sufficient to answer the question.

G. SELECTIVE-DECISION AND RECOVERY WORKFLOWS

Figure 3 provides a more detailed view of the selective-prediction and recovery procedures. The left panel follows an example from the frozen VQA model to the final decision. The model produces both an answer and a confidence score. Calibration-subset data are used to fit temperature parameters, refusal thresholds, and learned risk models. These choices are frozen before the report subset is ranked by native confidence or learned risk. The resulting ranking supports risk-coverage analysis and determines whether an example is answered or refused.

Accepted examples contribute to the answered subset and its associated risk. Refused examples are passed to the predicted defect head, which selects a likely failure mode and maps it to retake guidance. Recovery is therefore downstream of refusal: the defect head does not independently veto an answer that has already passed the confidence gate.

The right panel distinguishes the two recovery policies used in the evaluation. Under the ungated diagnostic policy, predicted defects are applied independently of the answer/refuse decision. This setting measures the intrinsic capacity of the defect head through ARR on unanswerable examples and the potential diagnostic burden through FRR on answerable examples. Under the refusal-gated policy, guidance is issued only for examples rejected by the confidence gate. Gated ARR and gated FRR therefore characterize recovery and burden within the subset that would actually receive retake instructions during deployment.

H. PROTOCOL SUMMARY

The complete methodology follows a consistent frozen-knob protocol. The visual backbones and VQA answers remain frozen. Answerability and defect heads are trained only on the training split. Thresholds, temperatures, and learned risk models are selected only on the calibration subset. All final measurements are produced on the held-out report subset. Results are aggregated across five random seeds where applicable, and the evaluation uses bootstrap confidence intervals, paired-bootstrap comparisons, and Benjamini–Hochberg false-discovery-rate correction for the relevant test families.

These choices define the scope of the contribution. The proposed system is not a new end-to-end VQA model. It is an attachable reliability layer that decomposes the assistive interaction into answerability triage, defect diagnosis, selective prediction, refusal explanation, and recovery. The methodology tests the contribution of each signal separately before measuring their learned combinations, allowing the value of auxiliary features to be reported as model-dependent rather than assumed to be universal.

VI. RESULTS

A. DATA SANITY

Taken together with the reliability-layer schematic in Figure 2 and the joined-label structure in Figure 1, the report-split evaluation starts from a deliberately difficult operating point rather than from a tuned end-to-end VQA system. The joined dataset contains 24,319 rows, with 20,000 training examples and 4,319 validation examples (1,295 calibration and 3,024 report rows). Over the full 24,319-row joined dataset, the frozen ViLT model obtains mean VizWiz answer accuracy 0.1853 (mean confidence 0.4469) and the frozen BLIP model obtains 0.1707 (mean confidence 0.3669); on the 3,024-row report split used for all selective-prediction results, ViLT reaches 0.2540 and BLIP reaches 0.2106. Both frozen answerers therefore leave substantial reliability headroom, which is exactly the setting we target: not building a stronger answerer, but quantifying when to trust, when to refuse, and how to recover around a fixed one.

B. ANSWERABILITY TRIAGE

Figure 4 and Table 1 show that answerability triage is useful but not sufficient on its own. CLIP performs best for answerability triage, reaching AUROC 0.7926 ± 0.0005 and AUPRC 0.8811. DINOv2 reaches AUROC 0.7769 ± 0.0005 , while MobileNetV3 reaches AUROC 0.7512 ± 0.0009 . All three exceed the majority-class F1 baseline of 0.809, with model F1 in the range 0.823–0.836, but the more conservative balanced-accuracy range of 0.59–0.66 shows that triage remains imperfect under class imbalance. Recent reliable-VQA work has shown that abstention quality depends not only on whether a signal correlates with error, but on how finely it orders risk across examples [6], [7]. Our result is consistent with that distinction: triage captures answerability structure,

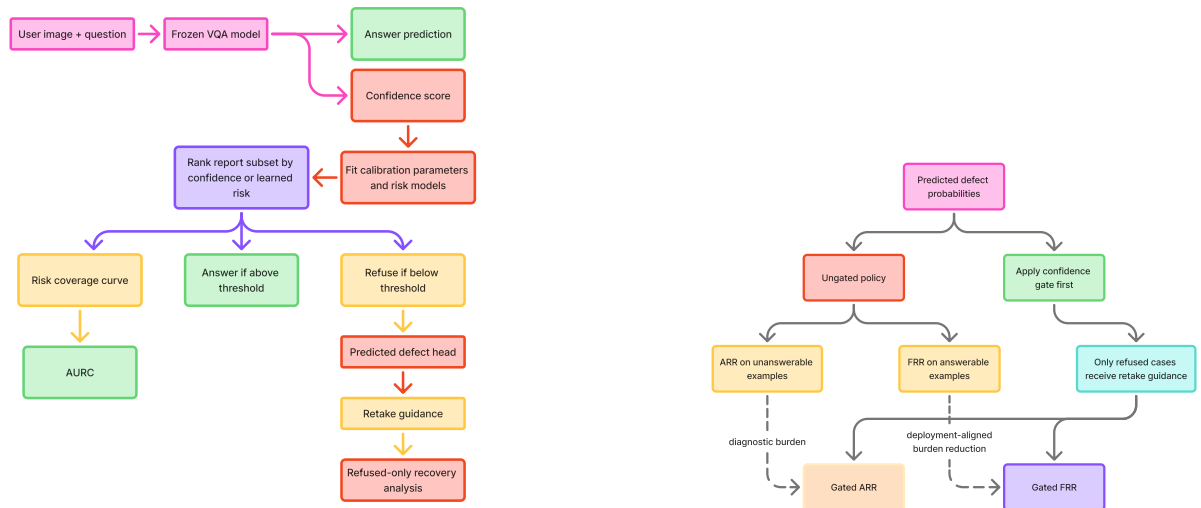


FIGURE 3. Methodological workflows for selective prediction and actionable recovery. Left: the frozen VQA model produces an answer and confidence score; calibration-subset data are used to fit calibration parameters, refusal thresholds, and learned risk models before the held-out report subset is ranked and divided into answered and refused examples. Refused examples receive predicted-defect explanations and retake guidance. Right: the ungated policy applies defect guidance independently of refusal and is evaluated using ARR and FRR, whereas the deployment-aligned gated policy issues guidance only after confidence-based refusal and is evaluated using gated ARR and gated FRR.

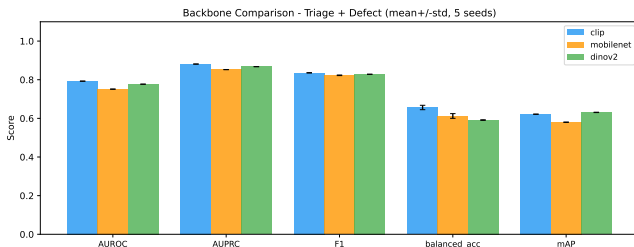


FIGURE 4. Backbone comparison for answerability triage and multi-label defect diagnosis. CLIP is strongest for answerability triage, while DINOv2 gives the highest defect mAP. Error bars show variation over five random seeds.

but it is not the same thing as a universally strong risk-ranking signal.

C. DEFECT DIAGNOSIS

For defect diagnosis, the backbone ordering changes. DINOv2 provides the strongest defect diagnosis by mAP, reaching 0.6309 ± 0.0009 . CLIP follows at 0.6216 ± 0.0007 , and MobileNetV3 reaches 0.5801 ± 0.0011 . Across backbones, unrecognizable content is the easiest defect to identify, with AUROC 0.93, while framing is the hardest by AUROC at 0.83. Under AUPRC, the rare defects—obstruction, bright, and dark—drop sharply to 0.33–0.48, making clear that high AUROC does not eliminate the burden imposed by rare-label retrieval. Figure 5 makes this visible at the per-defect level. This is where our reliability layer differs from selective-prediction work that stops at abstention: we explicitly pre-

serve the diagnostic signal even when its ranking value is weaker than confidence, because that signal is what supports explanation and retake guidance.

D. ACTIONABLE RECOVERY

The recovery results show why explanation should be evaluated separately from risk ranking. CLIP obtains the highest ungated actionable recovery rate, ARR 0.7196, followed by DINOv2 at 0.6907 and MobileNetV3 at 0.6278. The corresponding ungated false refusal rates are high, near 0.78. As noted in Section IV, this partly reflects the genuine prevalence of defects among answerable images; framing alone affects 56% of images, so a truthful defect detector can still generate a high burden proxy if it is decoupled from the answer/refuse decision. Figure 6 therefore should not be read as a deployment policy by itself. Instead, it supports our central role separation: defect diagnosis is inherently user-facing and action-oriented, not merely another way of sorting examples by confidence.

When retake guidance is gated by the refusal decision, the deployment picture becomes much more practical. At a 90% target coverage under the ViLT confidence gate, CLIP reaches gated ARR = 0.875 among refused unanswerable examples while reducing gated FRR to 0.050 on answerable examples. Under the BLIP gate, achieved coverage is 0.883 and CLIP reaches gated ARR = 0.794 with gated FRR = 0.102. Under the ViLT gate, the 80% setting yields achieved coverage 0.821, risk answered 0.698, refusal rate 0.179, and gated ARR/FRR of 0.809/0.120 for CLIP, 0.759/0.128 for

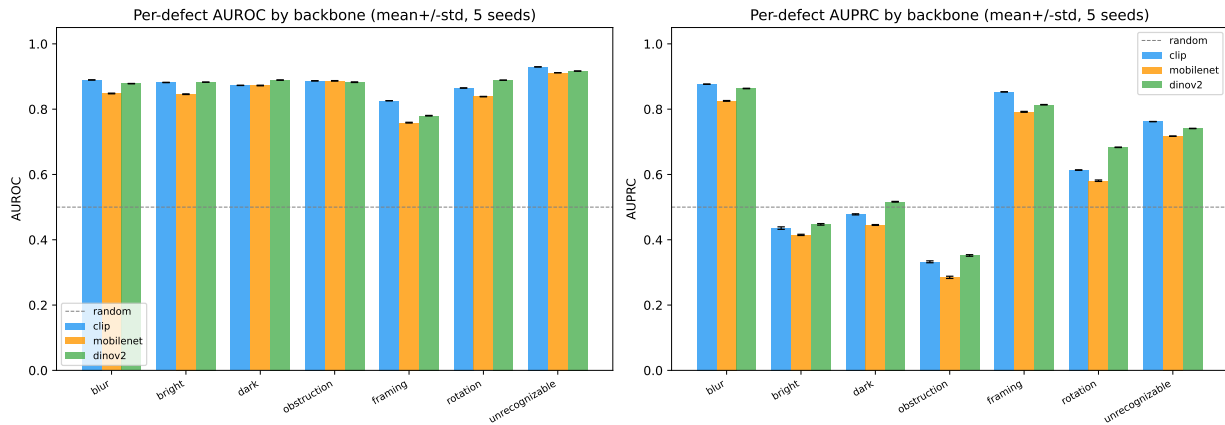


FIGURE 5. Per-defect diagnosis performance by backbone. AUROC is high for most defects, with framing the hardest label by AUROC. AUPRC exposes the effect of class imbalance: the rare defects (obstruction, bright, and dark) score far lower under AUPRC despite high AUROC.

TABLE 1. Core reliability-layer results. Triage AUROC and defect mAP are mean $\pm$ standard deviation over five seeds; ARR/FRR show bootstrap 95% confidence intervals. The final column (Δ AUROC, J–C) is the joint-head answerability AUROC minus that of the cascade design.

Backbone	Triage AUROC	Triage AUPRC	Defect mAP	ARR [95% CI]	FRR [95% CI]	Δ AUROC (J–C)
CLIP	0.7926 $\pm$ 0.0005	0.8811	0.6216 $\pm$ 0.0007	0.7196 [0.692, 0.750]	0.7746 [0.756, 0.793]	+0.0013
DINOv2	0.7769 $\pm$ 0.0005	0.8676	0.6309 $\pm$ 0.0009	0.6907 [0.660, 0.719]	0.7775 [0.760, 0.796]	+0.0028
MOBILENETV3	0.7512 $\pm$ 0.0009	0.8524	0.5801 $\pm$ 0.0011	0.6278 [0.597, 0.658]	0.7780 [0.760, 0.795]	+0.0037

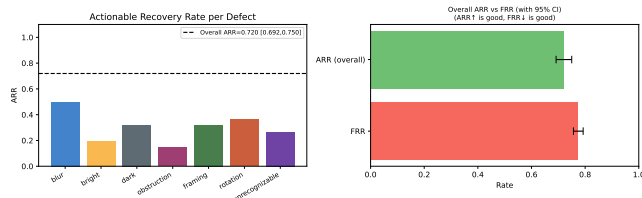


FIGURE 6. Actionable recovery from predicted defects. The left panel breaks down recovery by defect type; the right panel contrasts overall actionable recovery against the false refusal rate. The high false refusal rate underscores the need for cost-sensitive, refusal-gated recovery policies.

DINOv2, and 0.668/0.125 for MobileNetV3; the 70% setting yields achieved coverage 0.688, risk answered 0.657, refusal rate 0.312, and gated ARR/FRR of 0.758/0.217 for CLIP, 0.710/0.225 for DINOv2, and 0.637/0.222 for MobileNetV3. Table 2 shows that once recovery is conditioned on refusal, the defect head becomes a much more plausible assistive interface component. This is one of our main practical contributions: we evaluate recovery in a deployment-aligned refused-only regime rather than only as an ungated diagnostic.

E. SELECTIVE PREDICTION

The selective-prediction results are the most important negative finding in the paper, and they are also the place where over-generalization would be most misleading. For ViLT, global confidence is already the strongest risk-ranking signal we evaluated. It achieves AUC 0.5341 on the report split, whereas a random ordering yields 0.7460. Temperature scaling leaves the ranking unchanged and slightly worsens

expected calibration error on the report split, from 0.194 raw to 0.202 scaled. Defect-conditioned calibration produces small and statistically unsupported AUC changes ($p \geq 0.29$ across backbones), and learned confidence-plus-defect models are also unsupported ($p \geq 0.41$). Triage alone is much worse than ViLT confidence for all backbones, with improvement -0.176 to -0.185 and $p < 0.001$. Learned confidence-plus-triage and full-stack models produce only small gains that do not survive BH-FDR correction. Across all 22 ViLT selective-prediction tests in this family, no positive improvement over global confidence remains significant at FDR 0.05. This behavior is consistent with prior reliable-VQA findings that good abstention depends on the quality of the base correctness score, and it is also compatible with work showing that discriminative posteriors can remain reasonably calibratable even when raw confidence is imperfect [6], [7], [9], [10], [13], [35]. Our contribution here is the honest negative result: auxiliary visual signals did not beat a stronger discriminative confidence once correction for multiple testing was enforced.

The BLIP robustness analysis changes the interpretation without changing the standard of evidence. BLIP-VQA-base has report accuracy 0.2106 and a weaker global-confidence ranking, with AUC = 0.6242. In this setting, auxiliary reliability signals do help. With CLIP defects, a confidence-plus-predicted-defects model improves AUC by +0.0153, and the full stack improves AUC by +0.0179; both survive BH-FDR correction. With MobileNetV3 defects, the corresponding improvements are +0.0137 and +0.0172, also significant after correction. Confidence-plus-triage improves

TABLE 2. Refusal-gated recovery at calibration-selected confidence thresholds. Coverage and risk are shared across backbones because the refusal gate uses frozen VQA confidence; gated ARR and gated FRR depend on the defect head.

Backbone	Target coverage	Achieved coverage	Risk answered	Refusal rate	Gated ARR	Gated FRR
CLIP	90%	0.919	0.726	0.081	0.875	0.050
DINOv2	90%	0.919	0.726	0.081	0.817	0.055
MOBILENETV3	90%	0.919	0.726	0.081	0.769	0.054
CLIP	80%	0.821	0.698	0.179	0.809	0.120
DINOv2	80%	0.821	0.698	0.179	0.759	0.128
MOBILENETV3	80%	0.821	0.698	0.179	0.668	0.125
CLIP	70%	0.688	0.657	0.312	0.758	0.217
DINOv2	70%	0.688	0.657	0.312	0.710	0.225
MOBILENETV3	70%	0.688	0.657	0.312	0.637	0.222

BLIP risk ranking for all three backbones, from +0.0046 to +0.0069. DINOv2 full-stack reaches +0.0090 with $p = 0.043$, but does not survive FDR; triage alone remains much worse than confidence; and the ground-truth-defect oracle does not help BLIP, with -0.0007 and $p = 0.82$. The most careful interpretation is therefore not that “defects help selective prediction” in general, but that learned continuous auxiliary signals can improve a weaker base confidence signal. That reading is plausible in light of newer evidence that autoregressive or sequence-level confidence can be poorly calibrated at inference, that raw generation likelihoods can be weak proxies for answer quality, and that token-level or self-evaluation alternatives can sometimes correlate better with correctness [21], [36]–[38]. Our contribution is to show that this generative-confidence weakness manifests concretely in assistive VQA, but only for the weaker model and only under corrected testing.

Figures 7 and 8, together with Tables 3 and 4, make the role separation visually explicit. For ViLT, Figure 7 shows that temperature scaling does not materially alter the risk-coverage ordering, and Figure 8 shows that positive bars for auxiliary models are too small or too unstable to survive correction. For BLIP, the same diagnostic figure shows larger and corrected gains for predicted-defect and full-stack models, while the oracle result stays null. That oracle null matters: if ground-truth defect labels themselves do not improve ranking, then the useful signal is not a simple defect taxonomy ordering, but the richer continuous description encoded by the learned predicted signals. This is precisely where our paper adds something new to the literature: it separates explanatory value, recovery value, and ranking value instead of assuming that all three must rise and fall together.

F. QUALITATIVE REFUSAL BEHAVIOR

Figure 9 is useful because it shows why the paper should not be read as only a metric-improvement exercise. The grid is rule-sampled from the report split using fixed seed 7 rather than hand-picked, and its rows separate high-confidence correct answers, high-confidence wrong answers, good refusals with matching defect-based retake guidance, and wrong-reason refusals. The second row matters most

from an assistive perspective: visually fragile examples can still elicit confident but wrong answers, which is exactly the failure mode that motivates refusal. The bottom row shows the remaining limitation just as clearly: the system can refuse honestly yet still offer the wrong visual reason. Recent work on VQA reliability and multimodal uncertainty likewise emphasizes that identifying uncertainty is not the same as explaining it [6], [9], [38]. Our contribution is to place those two requirements in the same evaluation loop by measuring both whether the system should refuse and whether its explanation supports recovery.

G. JOINT VERSUS CASCADE

Finally, the architectural comparison shows that some design choices matter less than others. The joint head gives small but consistent AUROC gains over the cascade design: +0.0013 for CLIP, +0.0028 for DINOv2, and +0.0037 for MobileNetV3. These gains are real but modest, suggesting that backbone choice and base-confidence quality matter more than whether answerability and defect prediction are coupled in a single head. That synthesis matches the rest of the section: backbone quality affects triage and diagnosis, base VQA confidence determines whether auxiliary signals help ranking, and defect heads remain valuable for explanation and recovery even when their ranking gains are small. This is the practical conclusion our Results section is designed to support.

VII. DISCUSSION

Recent work on reliable VQA has shown that abstention can substantially improve system trustworthiness, but most of that literature evaluates success primarily through risk-coverage tradeoffs and selective-answering performance [6]–[10]. In assistive VQA, that framing is necessary but incomplete, because many failures originate not only in the answerer but also in the image capture process itself: VizWiz questions come from blind users, and a substantial fraction of images are genuinely hard to interpret because of blur, poor framing, occlusion, exposure problems, or unrecognizable content [1], [11]. Our results therefore support a sharper separation of roles than a pure abstention formulation

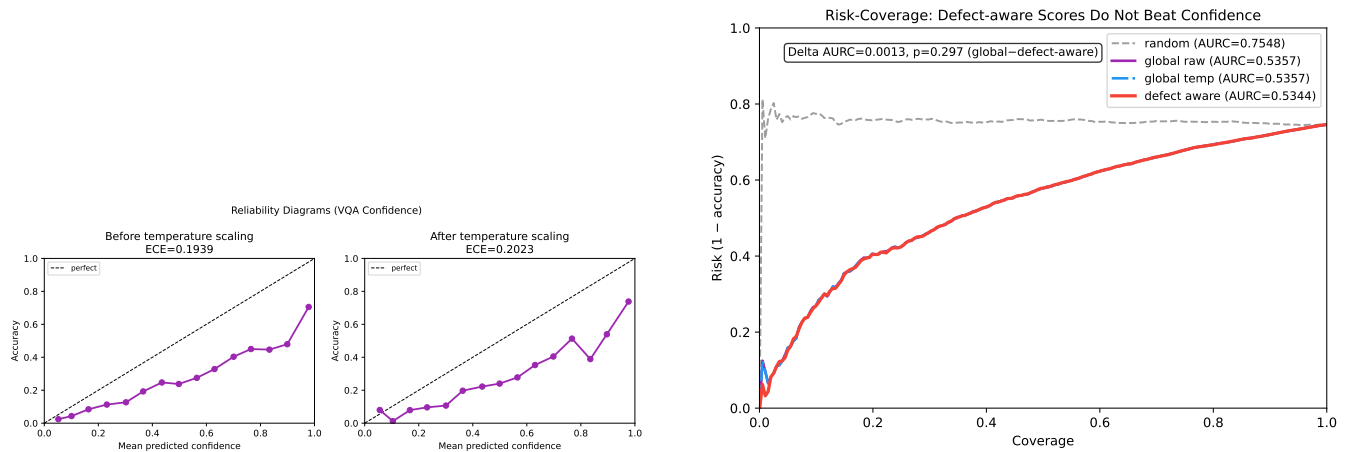


FIGURE 7. Calibration and selective-prediction behavior for frozen ViLT confidence. Temperature scaling does not close the reliability gap, and defect-aware risk scores do not improve the risk-coverage curve over ViLT global confidence.

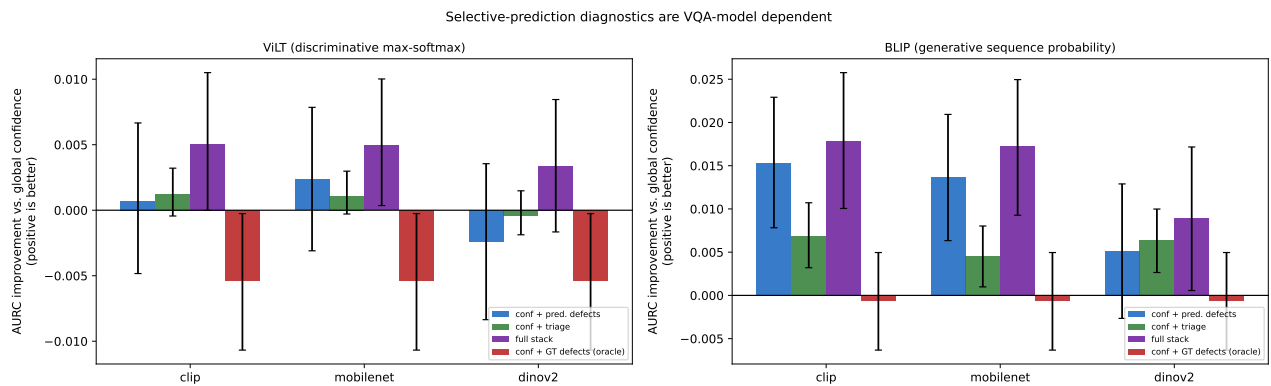


FIGURE 8. Selective-prediction diagnostics for two frozen VQA models. Positive values indicate lower AURC than each model's own global confidence. For ViLT, auxiliary signals do not produce an FDR-significant positive gain. For BLIP, learned predicted-defect and full-stack risk models improve the weaker generative confidence signal. Note the different y-axis ranges between the panels.

suggests: confidence is the primary candidate for ranking answer risk, while predicted defects contribute most reliably by explaining refusals and grounding recovery.

Why this separation holds strongly for ViLT but less strongly for BLIP is consistent with a broader calibration literature. In discriminative classification, posterior-style confidence is often imperfect yet still reasonably amenable to post-hoc calibration, and temperature scaling remains a strong baseline for converting logits into better-behaved confidence estimates [13], [35]. By contrast, confidence derived from autoregressive generation is often harder to interpret at the answer level: sequence probabilities are affected by tokenization, end-of-sequence behavior, decoding dynamics, and length normalization, and encoder-decoder models have been shown to exhibit substantial calibration problems even when token probabilities look locally sensible [39]. More broadly, several studies in language generation argue that raw sequence likelihood is an incomplete proxy for answer correctness because semantically equivalent responses can receive different probabilities, motivating answer-level self-evaluation or semantic uncertainty measures instead of re-

lying on token probability alone [40], [41]. Our ViLT/BLIP comparison fits this pattern: auxiliary triage and defect signals are not universally helpful, but they become useful when the base confidence signal is weaker as a risk ranker.

The BLIP results also clarify why predicted defects can help while the ground-truth-defect does not. Recent multimodal reliability work increasingly favors richer auxiliary signals—for example, consistency under perturbation, semantic agreement across responses, or explicitly calibrated verbal confidence—rather than assuming that a single symbolic attribute will linearly order risk [9], [42], [43]. That perspective aligns with our finding that the gain for BLIP comes from learned interactions with continuous predicted signals, not from defect labels as an oracle ordering. In other words, predicted defects appear to matter less as discrete declarations of “the” failure mode and more as soft descriptors of visual difficulty that can be fused with a weaker confidence signal. Our contribution is to demonstrate this distinction in an assistive VQA setting with human-labeled real-world defects: learned predicted signals can improve selective prediction for a weaker answerer, even when ground-truth defect

TABLE 3. ViLT selective-prediction diagnostics on the report split. Positive improvement means lower AURC than ViLT global confidence. BH-FDR correction is applied across the full family of 22 ViLT selective-prediction tests; the table reports the 16 tests of primary interest and omits six per-backbone confidence-recalibration variants. No positive improvement remains significant at FDR 0.05. The oracle model uses confidence plus ground-truth defects and is therefore backbone-independent; it is reported once.

Method	AURC improvement	<i>p</i>	BH-FDR result
Defect-conditioned calibration (CLIP)	+0.0009	0.297	n.s.
Defect-conditioned calibration (DINOv2)	−0.0001	0.923	n.s.
Defect-conditioned calibration (MOBILENETV3)	+0.0008	0.361	n.s.
Confidence + predicted defects (CLIP)	+0.0007	0.798	n.s.
Confidence + predicted defects (DINOv2)	−0.0024	0.448	n.s.
Confidence + predicted defects (MOBILENETV3)	+0.0024	0.418	n.s.
Triage score alone (CLIP)	−0.1836	< 0.001	worse
Triage score alone (DINOv2)	−0.1758	< 0.001	worse
Triage score alone (MOBILENETV3)	−0.1849	< 0.001	worse
Confidence + triage (CLIP)	+0.0012	0.168	n.s.
Confidence + triage (DINOv2)	−0.0004	0.604	n.s.
Confidence + triage (MOBILENETV3)	+0.0011	0.140	n.s.
Confidence + triage + predicted defects (CLIP)	+0.0051	0.053	n.s.
Confidence + triage + predicted defects (DINOv2)	+0.0034	0.205	n.s.
Confidence + triage + predicted defects (MOBILENETV3)	+0.0050	0.038	n.s.
Confidence + ground-truth defects (oracle)	−0.0053	0.039	n.s.

TABLE 4. BLIP selective-prediction diagnostics on the report split. Positive improvement means lower AURC than BLIP global confidence (AURC = 0.6242). BH-FDR correction is applied across the full family of 16 BLIP selective-prediction tests; the table omits three per-backbone confidence-recalibration variants. Seven positive improvements remain significant at FDR 0.05.

Method	AURC improvement	<i>p</i>	BH-FDR result
Confidence + predicted defects (CLIP)	+0.0153	< 0.001	better
Confidence + predicted defects (MOBILENETV3)	+0.0137	< 0.001	better
Confidence + predicted defects (DINOv2)	+0.0052	0.209	n.s.
Confidence + triage (CLIP)	+0.0069	< 0.001	better
Confidence + triage (MOBILENETV3)	+0.0046	0.008	better
Confidence + triage (DINOv2)	+0.0064	< 0.001	better
Confidence + triage + predicted defects (CLIP)	+0.0179	< 0.001	better
Confidence + triage + predicted defects (MOBILENETV3)	+0.0172	< 0.001	better
Confidence + triage + predicted defects (DINOv2)	+0.0090	0.043	n.s.
Triage score alone (CLIP/MOBILENETV3/DINOv2)	−0.137/ − 0.136/ − 0.130	< 0.001	worse
Confidence + ground-truth defects (oracle)	−0.0007	0.820	n.s.

labels by themselves do not.

The recovery results make the deployment implication more concrete. VizWiz images can remain answerable despite containing genuine quality defects [1], [11], so an ungated diagnosis module naturally produces a high refilm burden if it is allowed to recommend retakes on every example. That is why refusal-gated recovery is the right operating view for this setting: retake guidance should be issued only after the system has already decided that answering would be unsafe, so that diagnosis supports abstention instead of competing with it. Under that design, the point of defect prediction is not to replace confidence, but to make refusals more honest and more actionable. Our contribution is therefore not merely a better abstention score, but a deployment-aligned reliability layer that separates ranking, explanation, and recovery, and shows that the best balance among them

remains model-dependent.

VIII. LIMITATIONS

This study is limited to the examples obtained by joining the VizWiz-VQA and VizWiz-QualityIssues train and validation splits. The hidden VizWiz test answers are not available for evaluation, and the join retains only images for which both answerability and quality annotations are present. Consequently, the reported results characterize the shared, labeled portion of VizWiz rather than the complete VizWiz distribution, other assistive-image datasets, or images collected through a deployed application. Moreover, the calibration and report subsets originate from the same joined validation distribution. Although their separation prevents thresholds and risk models from being fitted on the reported examples, it does not test whether confidence, calibration, or refusal

thresholds remain reliable under changes in devices, users, environments, or question distributions. Predictive uncertainty and post-hoc calibration can degrade under distribution shift [44]; therefore, the present contribution should be interpreted as controlled within-dataset evidence for the proposed reliability layer rather than as a guarantee of deployment-time calibration.

The frozen VQA answerers are restricted to ViLT and BLIP-VQA-base, and both obtain low absolute accuracy on the joined data. The observed contrast between discriminative maximum-softmax confidence and generative length-normalized sequence confidence is therefore evidence of model-dependence within these two answerers, not proof that all discriminative and generative architectures will behave similarly. Accessibility-oriented VQA also differs materially from conventional machine-understanding benchmarks in its questions, image conditions, and user goals [45]. Recent evaluations focused on blind and low-vision (BLV) users further show that modern generative models can produce plausible but incorrect visual details, particularly when the submitted image is unanswerable [17]. Repeating the complete frozen-knob protocol with stronger contemporary vision-language models, alternative decoding procedures, and additional confidence estimators is therefore necessary before the model-dependent finding can be generalized beyond the two answerers evaluated here.

The answer-correctness target also inherits limitations from the standard VizWiz evaluation rule. Accuracy is computed from exact matches against the ten human reference answers using the VizWiz consensus score. This protocol provides a consistent basis for comparing all selectors in the present experiments, but it can penalize a generated answer that is semantically appropriate yet lexically different from the references. This concern is especially relevant for open-ended generative VQA, for which recent work has identified the inflexibility of exact-match evaluation and proposed semantic alternatives [46]. The risk labels used in our selective-prediction analysis should therefore be understood as correctness under the adopted VizWiz metric rather than as a complete measurement of usefulness, factual adequacy, or linguistic equivalence. Our contribution remains a controlled comparison because every risk-ranking method is evaluated using the same correctness definition, but future work should test whether the model-dependent conclusions persist under validated semantic and human evaluation.

The selective-prediction analysis examines confidence, temperature scaling, defect-conditioned calibration, answerability-triage scores, predicted-defect features, their learned combinations, and a ground-truth-defect oracle. Other signals—including answer consistency across stochastic decodes, representation-space uncertainty, question-conditioned features, retrieval evidence, or explicit self-evaluation—could produce different risk rankings. Five-seed reporting, paired bootstrap comparisons, and BH-FDR correction reduce sensitivity to initialization, sampling variation, and multiple testing within the specified experimental families, but they

do not establish that the observed ordering is invariant across datasets, VQA models, prompts, decoding strategies, or calibration populations. Accordingly, the study supports the narrower conclusion that the tested auxiliary signals are model-dependent under the specified protocol, rather than the broader claim that confidence must always dominate or that defect signals can never improve selective prediction.

The proposed ARR and FRR metrics evaluate whether a predicted defect corresponds to a predefined corrective action, but the deterministic defect-to-action mapping has not been validated with blind or low-vision users. A label-level recovery hit does not establish that the resulting instruction is clear, safe, contextually appropriate, or sufficiently precise to help the user capture a better image. Likewise, FRR measures the frequency with which an answerable example would trigger a retake suggestion; it does not directly measure frustration, physical effort, time, repeated capture attempts, or abandonment. The refusal-gated results provide a more deployment-aligned burden estimate than ungated diagnosis, but a longitudinal user study is still required to measure whether guidance improves the next image and whether users consider the intervention worthwhile. The rule-sampled qualitative grid avoids deliberate cherry-picking, yet it remains illustrative and cannot substitute for evaluation with the intended users. Thus, ARR and FRR should be viewed as reproducible offline proxies introduced by this work, not as complete measures of accessible user experience.

Finally, the seven quality labels explain important image-capture failures but do not cover every reason a VQA answer may be incorrect or unsafe. Errors can arise from ambiguous questions, optical-character-recognition failures, fine-grained recognition, small visual evidence, unsupported inference, missing world knowledge, or mismatch between the question and the visible content. The current system predicts image-level defect labels and does not identify the spatial region responsible for the answer or the refusal. Prior work on VizWiz answer grounding shows that localizing relevant evidence remains difficult, particularly when the evidence is small or the question requires text recognition [2]. Spatial grounding, region-specific camera instructions, question-aware diagnosis, and evaluation of successive retake attempts therefore remain future work. The present contribution is intentionally narrower: it establishes a lightweight image-level layer for deciding when to trust, explaining likely capture failures, and evaluating actionable recovery without claiming to resolve every source of VQA error.

IX. CONCLUSION

We have introduced a lightweight reliability layer for assistive VQA that leverages real answerability and image-quality labels from the VizWiz dataset to predict when a blind user's visual question should be answered or declined, and why. Small prediction heads on frozen visual encoders (CLIP, DINOv2, MobileNetV3) can effectively diagnose image defects (blur, lighting, framing, etc.) and forecast answerability, supporting guided retake instructions. In end-to-end evalua-

tion, these predictions enable high actionable recovery when used judiciously: for example, gating retake advice on a 90% coverage refusal threshold yields CLIP-gated ARR 0.875 at FRR 0.050 under ViLT.

At the same time, our selective-prediction analysis reveals that the utility of auxiliary signals depends on the VQA model. For the discriminative ViLT model (which chooses from a fixed vocabulary), its max-softmax confidence is already well-aligned with correctness, so defect and triage features provide no significant AURC gain after BH-FDR adjustment. By contrast, the generative BLIP model's length-normalized sequence confidence is less calibrated, and its risk ranking is significantly improved by learned combinations of confidence, triage, and predicted defects. In other words, augmenting confidence with auxiliary cues matters primarily when the base confidence is unreliable. This observation aligns with recent findings in VQA calibration: open-ended generation tends to be more overconfident than closed-set classification, and as Whitehead et al. showed, naively thresholding a model's softmax can limit coverage severely [6]. In our case, ViLT's base confidence needed almost no correction, whereas BLIP's did.

The practical lesson is thus to treat defect diagnosis as an explanation and fallback rather than a blanket substitute for confidence-based ranking. Start with the VQA model's confidence as the primary reliability signal, and only introduce learned auxiliary selectors when empirical calibration (on a held-out split) indicates that confidence is weak. When an answer is refused, the defect head can then give the user an interpretable reason and a concrete retake suggestion. This design neither overstates the power of defect labels nor discards them—it uses them exactly when and how they help. In future work, stronger vision-language models, richer language-based uncertainty measures, and real-world user studies should be explored to further validate and refine these reliability-layer strategies.

DATA AND CODE AVAILABILITY

Both datasets are publicly available from the VizWiz project [47], [48]. The full experiment pipeline (data assembly, training, calibration, selective-prediction diagnostics, recovery metrics, and figure generation), including cached-result manifests with git hashes for every reported number, is available at <https://github.com/meteorboyF/VQA-paper>.

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Reliability-layer behavior on rep examples (clip defect head)
(QUAL_SEED=7, sampled by rule - not hand-picked)

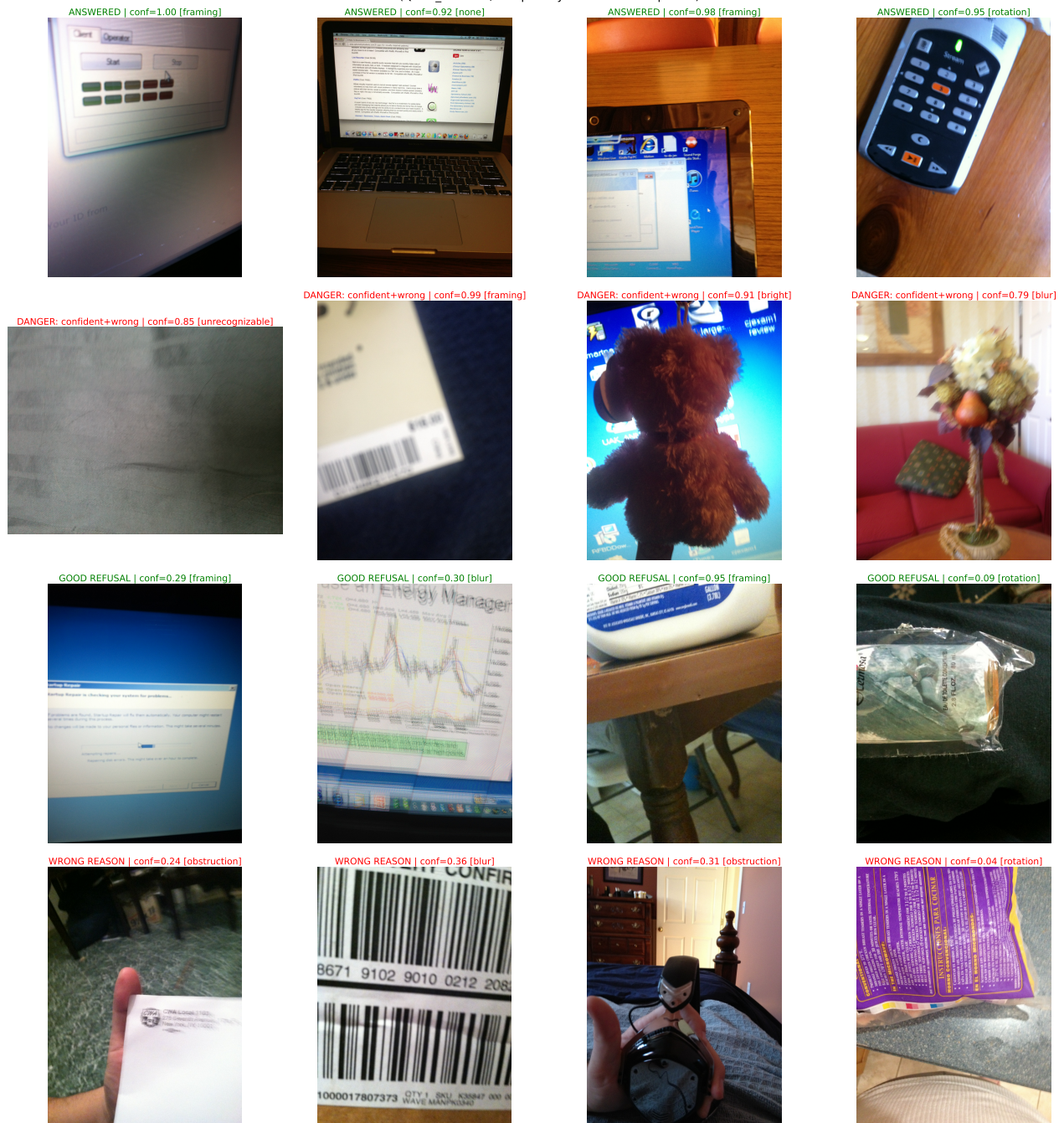


FIGURE 9. Rule-sampled qualitative examples from the report split (QUAL_SEED=7). Rows show high-confidence correct answers, high-confidence wrong answers, good refusals with matching defect-based retake guidance, and wrong-reason refusals. The figure illustrates both the motivation for refusal and the remaining challenge of explaining the correct visual failure mode.